LECTURE 1b / Fall 2026

Time Value of Money, Abstract Assets, and Net Present Value

CHEME 5660
Cornell University
August 27, 2026
Jeffrey D. Varner

LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

- **Convert dated dollars.** Explain why cash flows at different dates require common units and derive discrete and continuous discount factors.
- **Asset valuation.** Represent a project, product, or security as an abstract asset and compute its net present value.
- **Interpret the benchmark.** Explain what the sign of NPV does—and does not—say under a declared discount curve and cash-flow model.

1 Why Does Time Have a Price?

The time value of money begins with a simple asymmetry: one dollar in hand today and one dollar promised next year carry the same currency label, but they do not offer the same set of choices. Today's dollar can be consumed, held as a reserve, or invested. The future dollar cannot be used until it arrives, and its purchasing power and delivery are uncertain.

This idea predates modern financial markets. Irving Fisher organized it around two forces: impatience to spend income and the opportunity to invest it. Modern valuation adds inflation and risk to the same basic story. A discount rate is therefore not merely a computational input; it records the return forgone by waiting and, when appropriate, compensation for uncertainty.

L1a distinguished policy rates, bank funding rates, and asset-specific valuation rates. Here the discount rate is a declared model input: it describes the conversion between dated dollars under a stated benchmark and compounding convention. Monetary policy may influence that benchmark, but it does not uniquely determine the correct valuation rate for every asset.

2 Discounting as a Change of Units

A useful currency model treats money at different dates as different currencies: $\mathrm{USD}_0, \mathrm{USD}_1, \ldots$. Just as euros and U.S. dollars cannot be added before conversion, dated dollars cannot be added until they share a valuation date. The quoted yield specifies the conversion across time, and the accumulation and discount factors perform it. An abstract asset then provides a flexible model of the economic value of a process, good, or idea as dated signed cash flows (Fig. 1).

Start with one year of annual compounding. Suppose the initial principal P dollars earns the nominal annual yield y for one year. The resulting year-end future value F is given by Eq. 1:

$$F = (1 + y)P. \tag{1}$$

With annual compounding, the quoted yield y produces the forward conversion factor $1 + y$ and the backward

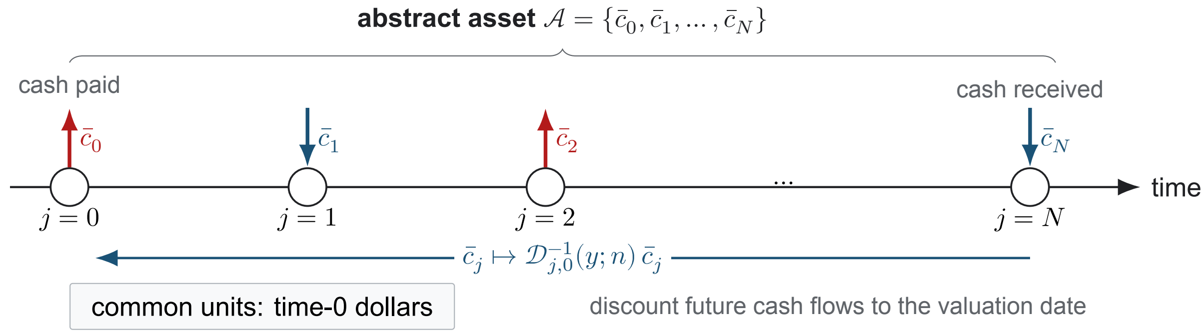

Figure 1: The abstract asset $\mathcal{A}$ has a signed net cash flow $\bar{c}_j$ at each index $j \in \{0, 1, \dots, N\}$, where $N = nT$ is the final cash-flow index. An arrow entering a timeline marker is money received; an arrow leaving it is money paid. The sign is already included in $\bar{c}_j$, so the labels need no extra minus sign. Discount each future cash flow back to the common valuation date $j = 0$.

factor $(1 + y)^{-1}$. These two directions define a reversible conversion between dated currencies (Defn. 2.1) and let us express every cash flow in Fig. 1 at a common date.

Definition 2.1. Accumulation and discount factors

Let time 0 be the valuation date, let the compounding-period index $j \geq 0$ identify a future date, and let USD_0 and USD_j denote dollars measured at those respective dates. For a quoted nominal annual yield y and a stated compounding convention, the *forward accumulation factor* $\mathcal{D}_{j,0}(y)$ is the multiplicative conversion factor from USD_0 to USD_j. If the present amount P_0 is measured in USD_0 and the equivalent future amount F_j is measured in USD_j, then $F_j = \mathcal{D}_{j,0}(y)P_0$ and $P_0 = \mathcal{D}^{-1}_{j,0}(y)F_j$, where $\mathcal{D}^{-1}_{j,0}(y) := 1/\mathcal{D}_{j,0}(y)$ is the *discount factor*. A valid pair satisfies $\mathcal{D}_{0,0} = 1$.
Throughout this course, $\mathcal{D}$ without an inverse always accumulates forward and $\mathcal{D}^{-1}$ always discounts back. Many finance texts use the phrase *discount factor* for both objects; the formulas are equivalent once the convention is stated, but we keep the two names distinct.

The periodic-compounding formula is a model rather than an algebraic identity. It assumes that the nominal annual yield y remains constant; the positive integer compounding frequency n divides each year into n equal intervals; interest is credited at the end of every interval at the proportional rate y/n; credited interest remains invested; and the investment horizon T years contains the whole number nT of intervals with no intervening deposits or withdrawals. That last assumption governs the accumulation of a single principal; it places no restriction on the cash-flow streams we discount in §3, which may have flows at any date.

Let the account value V_k denote the value immediately after interval k, and let the initial principal be $V_0 = P$. Applying the same one-interval update repeatedly gives the periodic accumulation factor:

$$
\begin{aligned}
V_0 &= P, && \text{initial deposit} \\
V_1 &= \left(1 + \frac{y}{n}\right) V_0 = P\left(1 + \frac{y}{n}\right), && \text{apply the interval multiplier once} \\
V_2 &= \left(1 + \frac{y}{n}\right) V_1 = P\left(1 + \frac{y}{n}\right)^2, && \text{apply it a second time} \\
&\vdots && \text{repeat the same update} \\
V_j &= P\left(1 + \frac{y}{n}\right)^j, && \text{after } j \text{ intervals} \\
\mathcal{D}_{j,0}(y; n) := \frac{V_j}{V_0} &= \left(1 + \frac{y}{n}\right)^j, && \text{normalize by the initial value.}
\end{aligned}
$$

The exponent j counts the compounding intervals and the base $1 + y/n$ is the one-interval multiplier, valid whenever $1 + y/n > 0$. A horizon of T years that lands on the compounding grid corresponds to $j = nT$, so we may write the same factor against elapsed time as $\mathcal{D}_{T,0}(y; n) = (1 + y/n)^{nT}$; the first subscript then carries years rather than a period count. As the compounding frequency n increases, this factor approaches the continuous-compounding limit (Thm. 2.1).

Theorem 2.1. Continuous-compounding limit

Let the nominal annual yield be the constant $y \in \mathbb{R}$, where $\mathbb{R}$ denotes the set of real numbers; let the investment horizon be $T \geq 0$ years; and let the positive integer n denote the number of compounding intervals per year, with $1 + y/n > 0$. As the compounding frequency n increases without bound, the discrete accumulation factor converges to its continuous counterpart, where e denotes Euler's number, as stated by Eq. 2:

$$\lim_{n\to\infty} \left(1 + \frac{y}{n}\right)^{nT} = e^{yT}. \tag{2}$$

Consequently, a continuously compounded rate g has forward accumulation factor $\mathcal{D}_{T,0}(g) = e^{gT}$ and discount factor $\mathcal{D}^{-1}_{T,0}(g) = e^{-gT}$.

Derivation. Apply the natural logarithm log to the discrete accumulation factor $\mathcal{D}_{T,0}(y; n)$ and use the Taylor expansion $\log(1 + x) = x - x^2/2 + O(x^3)$, where $x = y/n$ and $O(x^3)$ denotes a remainder bounded in magnitude by $C|x|^3$ near $x = 0$ for some constant $C > 0$:

$$\begin{aligned}\log \mathcal{D}_{T,0}(y; n) &= nT \log\left(1 + \frac{y}{n}\right)\\ &= yT - \frac{y^2T}{2n} + O(n^{-2}).\end{aligned}$$

Here $O(n^{-2})$ denotes terms bounded in magnitude by C'/n^2 for sufficiently large compounding frequency n and some constant $C' > 0$. These correction terms vanish as $n \to \infty$; exponentiating gives the result. The expansion also shows how quickly discrete compounding approaches the continuous convention. □

Thm. 2.1 is a statement about a limit, and it is easy to over-read. It does *not* say that a quoted nominal yield is already continuously compounded. For a *fixed* finite n, the exact continuously compounded equivalent of y is given by Eq. 3:

$$g_y := n \log\left(1 + \frac{y}{n}\right), \qquad \text{so that} \qquad \left(1 + \frac{y}{n}\right)^{nT} = e^{g_y T}. \tag{3}$$

Equation (3) is an identity, not an approximation: y with n compounding intervals and g_y compounded continuously are two spellings of the same accumulation rule. The limit in Eq. 2 instead compares y against a *different* rate as n grows, and the gap it closes is $O(y^2T/2n)$. Over a long horizon at a large yield that gap is not small: the companion notebook tabulates it and shows the leading-order estimate failing badly at $n = 1$.

Writing g for a continuously compounded rate also fixes a unit convention we use for the rest of the course: g carries units of inverse time, so the dimensionless log return accumulated over an interval Δt is $r = (\Delta t)\, g$. L2a uses this to compare a bill's cumulative gain r_B with its annualized growth g_B, and L3a uses it for equity log returns.

The companion notebook turns these expressions into computed factors. Use it to compare discrete and continuous compounding numerically and to see how quickly the two conventions converge.

COMPANION NOTEBOOK
L1b: Time Value of Money →
Build discrete and continuous discount factors and compare the two compounding conventions numerically.

3 Abstract Assets and Net Present Value

With the conversion rule in hand, we can return to the abstract asset. Let $\bar{c}_j$ be the signed net cash flow in USD_j: receipts are positive and payments are negative. An asset with $N+1$ cash-flow dates is represented by $\{\bar{c}_j\}_{j=0}^{N}$. NPV follows a simple discipline—*convert, then add*: multiply each $\bar{c}_j$ by $\mathcal{D}_{j,0}^{-1}(y;n)$ to express it in USD_0, then sum the converted values (Defn. 3.1).

Definition 3.1. Net present value

Let time 0 be the valuation date, let the nonnegative integer N be the terminal cash-flow index, and let each integer $j \in \{0, 1, \dots, N\}$ identify a cash-flow date. Let the signed net cash flow $\bar{c}_j$ be measured in USD_j, and let the accumulation factor $\mathcal{D}_{j,0}(y;n)$ convert USD_0 to USD_j under the nominal annual yield y compounded n times per year. Its reciprocal $\mathcal{D}_{j,0}^{-1}(y;n) := 1/\mathcal{D}_{j,0}(y;n)$ converts the cash flow back to time-0 dollars. The net present value at time 0 is given by Eq. 4:

$$\text{NPV}_0(y) = \sum_{j=0}^{N} \mathcal{D}_{j,0}^{-1}(y;n)\,\bar{c}_j = \left\langle \mathcal{D}_{\star,0}^{-1}(y;n), \bar{\mathbf{c}} \right\rangle, \tag{4}$$

where the $\star$ subscript denotes the ordered $(N+1)$-vector over $j = 0, \dots, N$. Every term is measured in time-0 dollars, so the sum produces one NPV value.

The sign of NPV compares the asset with the benchmark represented by the discount curve. The result depends on the stated cash flows and yield. It does not guarantee a realized profit or better performance than another investment:

Positive NPV ($\text{NPV} > 0$)**.** Discounted receipts exceed discounted payments. Under the stated cash-flow model and yield, the asset is worth more than the benchmark it is being priced against.

Zero NPV ($\text{NPV} = 0$)**.** Discounted receipts equal discounted payments, so the asset is fairly priced relative to that benchmark. Zero NPV does not mean zero nominal gain and does not mean zero risk.

Negative NPV ($\text{NPV} < 0$)**.** Discounted payments exceed discounted receipts, so the asset does not meet the required return used in the calculation. This does not by itself imply a nominal cash loss: undiscounted receipts may still exceed undiscounted payments once time is ignored.

The worked-example notebook applies the abstract-asset model to a vehicle purchase. It constructs the dated cash flows, builds the accumulation and discount factors, and computes the time-0 NPV as a scalar product.

COMPANION NOTEBOOK
L1b: Net Present Value Worked Example →
Represent a vehicle purchase as an abstract asset, build its discount factors, and compute the time-0 NPV.

KEY TAKEAWAYS

In this lecture, we placed dated cash flows in common time-0 units, derived discount factors, and defined NPV for an abstract asset.

- **Dated dollars require conversion.** Cash flows at different dates cannot be added economically until a discount factor expresses them at a common valuation date.
- **Compounding is a convention.** A nominal yield y compounded n times per year and its continuous equivalent $g_y = n\log(1+y/n)$ are the same rule written two ways; neither should be used without stating its assumptions.
- **Structure before identity.** An abstract asset describes any investment as a dated sequence of payments and receipts.

Next, use the companion notebooks to build these factors numerically and compute an NPV end to end.

ADDITIONAL READING

- Irving Fisher, *The Theory of Interest* (1930), Chapters I–III, for the classical account of impatience and investment opportunity.
- Jonathan Berk and Peter DeMarzo, *Corporate Finance*, 5th ed. (2020), Chapters 3–4, for present value and investment decision rules.
- Frank J. Fabozzi and Francesco A. Fabozzi, *Bond Markets, Analysis, and Strategies*, 10th ed. (2021), for discount factors and fixed-income conventions.

Educational use and risk notice. These notes are for instruction only and do not constitute investment advice. Financial decisions involve risk, and model outputs depend on assumptions.